

AMAN SHRESTHA

Backend Engineer · Python · Real-Time Systems · API Design
Baltimore, MD · amanshrestha3003@gmail.com · github.com/AmanShrestha-01 · linkedin.com/in/aman-shrestha

EDUCATION

Morgan State University Baltimore, MD · May 2028
Bachelor of Science in Computer Science
▶ Coursework: Data Structures & Algorithms, Systems Programming (C/C++), Database Systems, Computer Architecture, OOP

TECHNICAL SKILLS

Languages	Python, Java, C/C++, JavaScript, SQL
Frameworks	Flask, Flask-SocketIO, SQLAlchemy, Pytest
Databases	PostgreSQL, Redis, SQLite
Infrastructure	Docker, Docker Compose, Render, GitHub Actions CI/CD
Tools	Git, REST APIs, JWT Auth, Bcrypt, Stripe API, WebSockets, Swagger/OpenAPI

PROJECTS

Real-Time Chat Service Live · 2025
Flask · Flask-SocketIO · Redis Pub/Sub · PostgreSQL · Docker Compose

- ▶ Built a WebSocket chat server supporting multiple concurrent rooms, live presence tracking, and persistent message history in PostgreSQL
- ▶ Used Redis Pub/Sub to broadcast messages across server instances — enables horizontal scaling without pinning users to one server

E-Commerce Platform Live · 2025
Flask · PostgreSQL · SQLAlchemy · JWT · Stripe · Swagger · Pytest · Render

- ▶ Designed a 20+ endpoint REST API covering the full purchase flow: product catalog, cart management, order processing, and Stripe payments
- ▶ Role-based access control (customer vs. admin) with JWT auth; full Pytest suite covering auth, Stripe edge cases, and input validation

AI-NutriPlan — Nutrition Tracking API Live · 2025
Flask · SQLAlchemy · SQLite · Claude AI · JWT · Swagger

- ▶ Meal-tracking API where users set macro goals, log daily intake, and receive AI-generated meal plans personalized to their targets via Claude

AI Study Assistant API Live · 2025
Flask · SQLAlchemy · Claude AI · JWT · Swagger

- ▶ Processes uploaded notes and uses Claude to generate summaries, quiz questions, and study guides; per-user rate limits to manage AI costs

Bookmarks REST API Live · 2025
Flask · SQLAlchemy · JWT · Bcrypt · Render

- ▶ CRUD API with signup, Bcrypt hashing, and JWT auth — bookmarks isolated at the query level so users can only access their own data

EXPERIENCE

Backend Engineer · Self-Directed Remote · Jan 2026 – Present
▶ Building production-grade backend APIs through a structured engineering program — five projects architected, tested, and deployed to production

Web & Tech Support Intern · Valentino's Restaurant Baltimore, MD · Jun – Aug 2025
▶ Maintained restaurant web systems and managed backend content via CMS; diagnosed and resolved technical issues to improve site reliability

Restaurant Tech & Web Assistant · Venetian Italian Eatery Edgewood, MD · Dec 2024 – Feb 2025
▶ Managed website content and structured menu/pricing data via CMS; configured Toast POS systems for order processing and payments

ADDITIONAL

- ▶ All projects publicly deployed with live URLs, Swagger documentation, and clean commit history on GitHub
- ▶ Interests: Systems design, backend architecture, high-performance psychology, soccer